Narayan Sahu

Data Engineer | GenAI | Databricks | Snowflake | Azure

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Summary

Data Engineer with 5 years of experience architecting scalable ETL/ELT pipelines, real-time data streaming, and Lakehouse solutions within the US Healthcare domain. Proven success delivering mission-critical Medallion architectures for UnitedHealth Group using Azure Data Factory, Databricks, Snowflake, PySpark, and Apache Airflow. Deep business domain expertise in Medicare/Medicaid enrollment, billing, claims adjudication, and Health Risk Assessments (HRA). Expanded expertise into LLMOps and AI-enabled workflows, successfully deploying production-ready RAG-based GenAI chatbots to accelerate healthcare clinical insights and operational efficiency.

Technical & Domain Skills

Languages: Python, SQL, PySpark

Cloud Platforms: Azure (Data Factory, Event Hubs, Blob Storage, Fabric), AWS S3

Data Engineering: ETL/ELT, Databricks (Auto Loader, Jobs), Apache Airflow, ADF, SSIS, Kafka, Live Data Streaming

Data Warehousing: Snowflake (Architecture, Snowpipe), Azure Synapse, Delta Lake, Medallion Architecture

GenAI & LLMOps: Retrieval-Augmented Generation (RAG), Azure OpenAI, Vector Databases, Semantic Search, LangChain, AI DLC (Discovery, Inception, Construction Readiness, Operations for Greenfield & Brownfield)

Healthcare Domain: Member Enrollment, Claims Processing, Medical Billing, Health Risk Assessments (HRA), D-SNP

DataOps & CI/CD: Git, GitHub Actions, Azure DevOps, Data Quality Validation, Pipeline Orchestration

Work Experience

Optum India (UnitedHealth Group), Bangalore, India April 2024 - Present

Data Engineer

- Designed and optimized ETL/ELT Medallion pipelines using Azure Data Factory (ADF), Databricks, and Python, improving daily load performance by **30%**.
- Implemented rigorous data quality checks and governance rules using Python and SQL, increasing data reliability for downstream Power BI reporting by **95%**.
- Integrated and processed multi-source structured and unstructured data from Snowflake, Cosmos DB, and Blob Storage, handling high-volume daily ingests of US Healthcare member enrollment, claims, and provider demographic data.
- Engineered production-ready RAG chatbot solutions leveraging Vector DBs and LLMs to answer complex business and member eligibility queries, streamlining billing and HRA compliance processes.
- Orchestrated decoupled workflows leveraging Apache Airflow DAGs and ADF triggers, standardizing CI/CD deployments and reducing manual intervention by **90%**.

Optum India (UnitedHealth Group), Bangalore, India Jan 2022 - March 2024

Associate Data Engineer

- Developed robust ETL pipelines using SSIS and Python (Pandas, NumPy) to support critical state verification processes, Medicaid enrollment (D-SNP), and claims adjudication workflows.
- Built complex SQL views, stored procedures, and triggers for business logic; improved query execution times by **25%** through indexing and optimization.
- Migrated legacy on-prem SSIS workflows to Azure using ADF, ADLS Gen2, and Snowflake, improving scalability and significantly reducing compute costs.
- Implemented automated data validation scripts and testing protocols to ensure data consistency across development and production environments.
- Collaborated in PODs and Agile/Scrum ceremonies and actively supported DataOps deployment phases to ensure seamless release cycles.

Projects

HRA Data Mart & Healthcare GenAI Assistant Optum India | Mar 2025 - Present

Databricks • Delta Lake • Airflow • Kafka • Azure Blob • RAG / LLM

- Engineered an end-to-end Medallion pipeline handling daily Health Risk Assessment (HRA) ingestion across Medicare Advantage (MA) channels using **Databricks Auto Loader and Jobs**.
- Orchestrated decoupled data ingestion workflows in **Apache Airflow** and ADF, delivering cleansed assessment scores to downstream claims processing systems within statutory 90-day windows.
- Designed and deployed an internal **RAG-based conversational AI system** using vector embeddings to allow clinical and business teams to query member HRA statuses in natural language, reducing manual ticket lookups by **60%**.

Next-Gen Unified Member Data Hub (GPS Engine) Optum India | April 2024 - March 2025

ADF • Snowflake • Azure Event Hubs • Cosmos DB • Oracle • Power BI

- Architected a multi-source Medallion Lakehouse unifying unstructured file feeds, relational Oracle snapshots, and real-time CDC streams from **Cosmos DB via Azure Event Hubs**.
- Implemented **ADF pipelines** for scalable transformations and semantic models on Snowflake, replacing bulky stored procedures and reducing model execution time by **35%**.
- Created standardized fact and dimension structures feeding **Power BI dashboards** for member demographic distribution, billing, and claims analytics, enforcing strict row-level security.

Dual Special Needs Plan Member Eligibility System (DMEVS) Optum India | Jan 2022 - March 2024

ADF • Snowflake • SSIS (Migration) • SQL • Medallion Pipeline

- Led legacy modernization of SSIS pipelines to **Azure Data Factory (ADF)**, reducing daily enrollment and eligibility extraction latency by **45%** for multi-state D-SNP programs.
- Designed complex eligibility rule calculation engines and payer high-risk trend models in **Snowflake**, processing millions of member records with SCD Type 2 tracking for annual reverifications.
- Built rigorous automated data quality validation gates checking coverage overlaps, demographic variances, and state compliance prior to syncing downstream delta changes into the BASIS system.

Education

Master in Computer Application 2018 - 2021

Fakir Mohan University, Balasore ***CGPA: 8.2/10***

Bachelor of Computer Applications (BCA) 2015 - 2018

Berhampur University, Bhubaneswar ***70%***

Additional Highlights

- Strong understanding of SDLC, POD-based delivery models, Agile/SCRUM, and Test-Driven Development (TDD)
- Excellent communication, problem-solving, and team collaboration skills
- Quick learner with a proactive approach to new technologies and challenges
- Certified through Optum AI Dojo program and recognized among the Super 30 by Optum's CTO.